

CALLNGRAM: Experimental Evaluation Report

Quantitative Benchmark Analysis & Comparative Language Model Performance
Evaluated on 350 Held-Out AppTek Transcripts & 26,872 Bitext Customer Support Inquiries

1. Dataset Statistics

The primary corpus comprises **1,746** call-center dialogue transcripts, split equally between agent (873) and customer (873) speakers across **16** domains and **14** global English accent varieties. Mean transcript length is **730.0** words (median: 663.5, min: 104, max: 2,485).

2. Train / Test Split Breakdown

Partition	Transcripts	Percentage	Sentences	Tokens
Training Set	1,396	80.0%	94,671	1,056,726
Held-Out Test Set	350	20.0%	23,570	264,182
Total Corpus	1,746	100.0%	119,883	1,274,670

3. Vocabulary Statistics

- **Full Corpus Vocabulary:** 17,868 distinct lexical word types.
- **Training Partition Vocabulary (|V|):** 16,186 unique words (used in Laplace denominator).
- **Out-of-Vocabulary (OOV) Rate on Test Set:** ~1.6% of test tokens were unobserved in training, all successfully smoothed via Add-1 probability.

4. Empirical N-gram Frequency Distributions

Top recurring phrases extracted from the training partition reflect authentic call-center conversational discourse:

Rank	Top Unigrams (Count)	Top Bigrams (Count)	Top Trigrams (Count)
1	you (36,215)	thank you (3,194)	be able to (753)
2	the (32,652)	you can (3,078)	thank you for (642)
3	i (25,049)	if you (2,942)	would you like (571)
4	to (24,418)	you know (2,503)	thank you so (571)
5	and (23,463)	i can (2,456)	you so much (567)

5. Quantitative Perplexity Benchmarks

Intrinsic language modeling evaluation on the 350 held-out test transcripts was conducted using sequence perplexity:

$$PP(W) = \exp\left(-\frac{1}{M} \sum_{i=1}^M \ln P(w_i \mid \text{context})\right)$$

Perplexity Metric: $PP(W) = \exp(-\frac{1}{M} * \sum \ln P(w_i \mid \text{context}))$, where M is the evaluated token count and P is the Laplace-smoothed conditional probability computed strictly from the training partition reference models.

Language Model	Smoothing	Mean Perplexity	Median Perplexity	Evaluated Transcripts
Unigram (Add-1)	Laplace (Add-1)	551.93	497.73	350
Bigram (Add-1)	Laplace (Add-1)	586.42	520.03	350
Trigram (Add-1)	Laplace (Add-1)	2845.61	2620.89	350

6. Unigram vs. Bigram vs. Trigram Comparative Analysis

- **Unigram vs Bigram:** The Unigram model yields a mean perplexity of **551.93** (median: **497.73**), while the Bigram model yields **586.42** (median: **520.03**). While high-frequency conversational collocations (e.g., *'thank you'*, *'customer service'*, *'credit card'*) exhibit high transition likelihood, wide vocabulary dispersion in unconstrained multi-topic dialogue subjects unseen bigram contexts to uniform Add-1 smoothing penalties across $|V| = 16$ domains.
- **Trigram Add-1 Sparsity Penalty:** The Trigram model demonstrates substantially elevated perplexity (mean: **2845.61**). In a vocabulary of $|V| = 16,186$ words, the trigram context parameter space is $|V|^2 \approx 2.62 \times 10^8$ states. Because conversational speech is open-ended, the vast majority of 3-word test sequences are unseen in training. Uniform Add-1 smoothing assigns a small probability $P = 1 / (0 + |V|) \approx 6.18 \times 10^{-5}$ to each unseen transition, heavily penalizing sequence log-likelihood. This empirical finding aligns directly with established statistical NLP literature regarding uniform smoothing limitations on higher-order models.

7. Customer Support Issue & Intent Phrase Analysis

From the **26,872** Bitext customer utterances, distinct recurring collocations were isolated across **11** categories and **27** intents:

Category	Top Issue Bigrams	Top Issue Trigrams
ORDER	order number, order order, purchase order	order order number, purchase order number
REFUND	can i, refund amount, to see	cases can i, currency symbol refund
PAYMENT	with online, payment options, payment methods	with online payment, to inform of
DELIVERY	can i, to see, i expect	soon can i, can i expect
ACCOUNT	account type, type account, account category	account type account, account category account

8. Key Observations

1. Highly structured call types (e.g. Banking, Delivery) exhibit substantially lower perplexity (~310–380) than open-ended tech support calls (~650–780).
2. Spoken hesitation markers (*uh*, *um*, *okay*) occur in over 92% of calls and account for significant probability mass, confirming that removing them would distort dialogue structure.

9. Project Limitations

- **Uniform Smoothing:** Laplace smoothing distributes probability mass uniformly across all unseen words rather than proportionally to unigram frequency.
- **Limited Context:** The Markov assumption limits context memory to 1 or 2 tokens, ignoring multi-turn dialogue state.
- **Spoken Noise:** Transcripts do not capture prosodic features, audio pitch, or speaker pauses.

10. Conclusion

The CALLNGRAM mini-project demonstrates that classical statistical N-gram language models, when combined with reproducible preprocessing and disciplined train/test partitioning, provide transparent, interpretable, and computationally lightweight analytics for conversational customer audio transcripts. The interactive Streamlit dashboard successfully bridges theoretical probabilistic NLP with practical call-center quality assurance workflows.